Before digging into quantum gravity proper, we need to lay some groundwork. General relativity is a theory of the dynamics of space-time, so in this chapter we'll ask why the concept of "space" is so important in the first place. The answer resides in the concept of *locality*—things interact with one another when they are nearby in space. In the next chapter we'll see how quantum fields propagating through space embody this principle of locality, and teach us something about the nature of empty space. In the chapter after that we'll investigate how to extract space itself from the quantum wave function. And in the final chapter we'll see that when gravity becomes strong, locality itself will have to be abandoned as a central principle. The mystery of quantum gravity seems to be intimately connected with the virtues and the shortcomings of the idea of locality.

It's worth being careful about "locality," as it is used in two somewhat different senses: what we might call *measurement locality* and *dynamical locality*. The EPR thought experiment shows that there is something that seems nonlocal about quantum measurement. Alice measures her spin, and what Bob will measure for his spin far away is immediately affected, even if he doesn't know it. Bell's theorem implies that any theory in which measurements have definite outcomes—basically, every approach to quantum mechanics other than Many-Worlds—is going to feature this kind of measurement nonlocality. Whether Many-Worlds is nonlocal in this sense depends on how we choose to define our branches of the wave function; we're allowed to make either local or nonlocal choices, where branching happens only nearby or immediately all throughout space.

Dynamical locality, on the other hand, refers to the smooth evolution of the quantum state when no measurement or branching is happening. That's the context in which physicists expect everything to be perfectly local, with disturbances at one location only immediately affecting things right nearby. This kind of locality is enforced by the rule in special relativity that nothing can travel faster than light. And it's this dynamical locality that we're concerned with at the moment as we study the nature and emergence of space itself.

With that in mind, we can roll up our sleeves a bit and dig into the question of how the structure of our observed reality—we live in a world that looks like a collection of objects located in space, behaving approximately classically except for occasional quantum jumps—emerges from the quantum wave function. Everettian quantum mechanics purports